Data Mining & Machine Learning
M.Sc. in Artificial Intelligence and Data Engineering
University of Pisa

Project Report

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Project Title: Loan Default Prediction

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Abstract

This work addresses the problem of credit default prediction as a binary classification task over a loan dataset comprising approximately 148 000 records and 34 features (including target). The dataset presents a significant class imbalance, with the positive class (default) being about 25%, which motivates the adoption of evaluation metrics robust to skewed distributions, namely AUC, balanced accuracy, recall, and F1-score.

Six classifiers were evaluated: Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, AdaBoost, and Gaussian Naive Bayes. Each model was embedded in a scikit-learn pipeline incorporating median imputation and standardisation for numerical features, and mode imputation with one-hot encoding for categorical ones. Model selection and performance estimation were conducted through a nested cross-validation scheme combining randomised and local grid search in the inner loop, with a five-fold stratified split in the outer loop, ensuring unbiased generalisation estimates and preventing data leakage.

Random Forest and Decision Tree emerged as the top-performing models, achieving AUC scores of 0.866 and 0.856 and balanced accuracies of 0.792 and 0.785 respectively. A pairwise Wilcoxon signed-rank test found no statistically significant difference between the two ($p = 0.0625$), leading to the selection of the Decision Tree on grounds of parsimony and interpretability. The final model was subsequently re-tuned on the full dataset and analysed through SHAP and LIME explainability frameworks, identifying credit score and debt-to-income ratio as the most influential predictors of default.

1. Introduction

Credit risk assessment is a fundamental task in the banking and financial industry. The consequences of poor credit risk management are significant on multiple fronts: undetected defaults translate into direct financial losses for lending institutions, while overly conservative models may deny credit to solvent customers, reducing business opportunities. Traditional credit scoring methods, such as logistic regression-based scorecards, have long been the industry standard [1]; indeed, logistic regression remains a benchmark in the credit risk industry, largely because the limited interpretability of more complex models is difficult to reconcile with the requirements of financial regulators [2]. However, the growing availability of data has motivated the adoption of supervised machine learning techniques, which are better equipped to capture complex, non-linear interactions among financial and demographic variables and have been shown to outperform traditional scorecards in large-scale benchmarks [3].

The dataset used in this project is characterised by a mix of numerical and categorical features and a substantial class imbalance between default and non-default cases, reflecting the real-world distribution of credit events. These properties make it a representative and challenging benchmark for classification methods.

This project aims to:

1. Train and compare a diverse set of classifiers spanning different learning paradigms linear models, decision trees, bagging and boosting ensembles, and probabilistic models in order to assess which inductive bias best captures the predictive structure of the data.

2. Evaluate model performance under class imbalance by adopting metrics that reflect the asymmetric costs of misclassification in the credit domain: a false negative (an undetected default) represents a direct financial loss for the institution, while a false positive (a solvent customer incorrectly rejected) entails lost revenue and potential reputational damage. Recall and balanced accuracy are therefore treated as primary objectives, without disregarding precision.
3. Provide both global and local model explanations using Explainable AI (XAI) techniques, specifically SHAP and LIME, in order to identify the features that drive the model's decisions at the population level and to interpret individual predictions on specific records, supporting transparency and trust in the deployed model.

References

[1] *Benchmarking state-of-the-art classification algorithms for credit scoring*. Wiley.

[2] *Machine learning for credit scoring: Improving logistic regression with non-linear decision-tree effects*. Wiley.

[3] *Benchmarking state-of-the-art classification algorithms for credit scoring: An update of research*. Wiley.

2. Problem

Table 1 summarizes the problem and dataset.

Table 1. Problem and dataset at a glance.

Attribute	Summary
Task type	Binary classification
Dataset	Loan Default Dataset, 148.000 and 34 features
Target variable	Status, indicating future loan default (0: no default, 1: default)
Features	34 features: 13 numerical, 21 categorical and target see Section 2.1
Missing values	26.66% Upfront charges, 24.64% IRS, 24.51% ROI, 16.22% dtir1, 10.16% LTV and prop value and other for a cumulative 10%
Class distribution	Default (1): 25% Not default (0): 75%, unbalanced
Evaluation metric(s)	AUC, Recall, balanced accuracy and F1

The problem is binary classification on an unbalanced dataset with many missing values. The metrics used for evaluation are coherent with the target balance.

2.1 Exploratory Data Analysis

The target distribution is markedly imbalanced: 112,031 records (75.36%) belong to the negative class (no default) and 36,639 records (24.64%) to the positive class (default), yielding an imbalance ratio of roughly 3:1..

Most numerical features exhibit right skewed distributions, like `income` or `loan_amount`, while others, such as `credit_score` and `debt_to_income_ratio`, are more symmetrically distributed in both classes.

The Pearson correlation heatmap of numerical features (Figure 1) shows weak values, especially with the target variable, still some high correlation appear between features that are logically related in the banking domain.

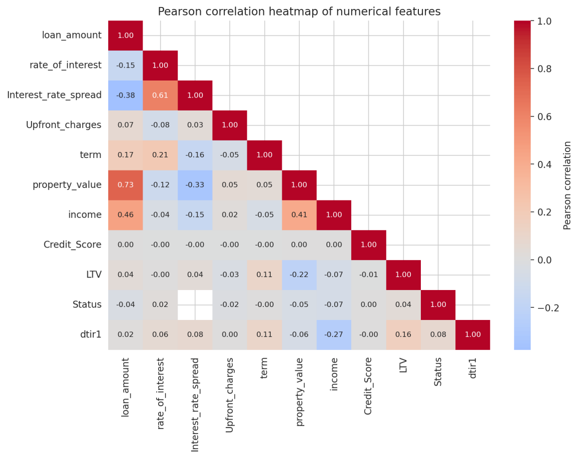

Figure 1. Pearson correlation heatmap of the numerical features (lower triangle).

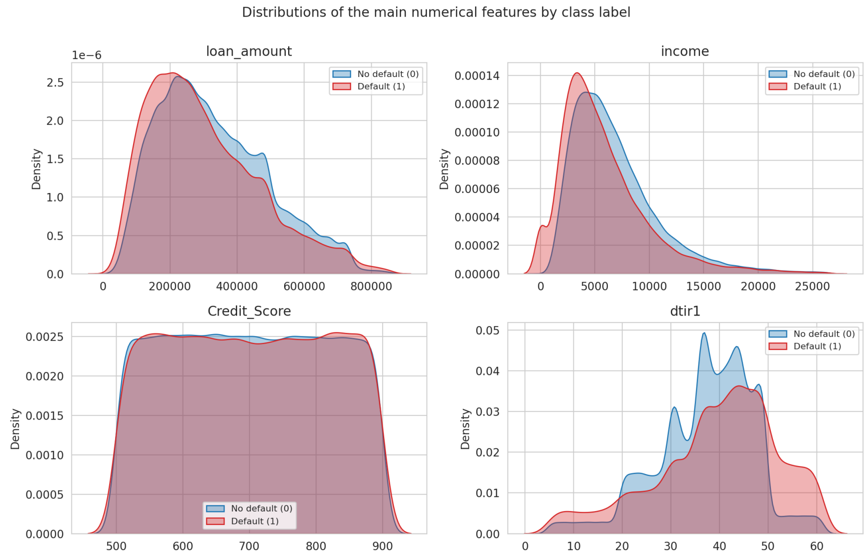

Figure 2. Feature distributions by class label.

Fourteen features contain missing values. The overall missingness ranges from negligible (0.03% for `term`) to substantial (26.66% for `Upfront_charges`). For a group of price related features missingness is aligned with target, `Interest_rate_spread` is missing for 100% of the defaulted loans and for 0% of the repaid ones; `rate_of_interest` and `Upfront_charges` follow the same pattern (99.45% and 99.58% missing in the positive class against 0% and 2.82% in the negative class.

2.2 Feature Description

Table 2. Dataset attributes. Type: N = numerical, C = categorical (B = binary, O = ordinal). † = anonymized levels.

Feature	Type	Meaning	Why informative
ID	N	Unique application identifier.	None (identifier).
year	N	Application year (constant, 2019).	None (zero variance).
Gender	C	Male, Female, Joint, or not available.	General info.
age	C (O)	Age bracket, 7 bins ($<$25 to $>$74).	Income stability, credit history length.
income	N	Declared monthly income.	Repayment capacity.
Region	C	Property region (4 areas).	Regional economic conditions.

Table 2 (continued)

Feature	Type	Meaning	Why informative
business_or_commercial	C (B)	Business vs. personal purpose.	Business loans are riskier.
loan_limit	C (B)	Conforming (cf) vs. non-conforming (ncf).	Non-conforming = higher risk.
loan_type†	C	Loan product (type1–3).	Product-specific risk.
loan_purpose†	C	Purpose (p1–p4).	Classic risk driver.
loan_amount	N	Principal (16.5K–3.58M).	Exposure size vs. income.
term	N	Duration in months (96–360).	Longer horizon, more uncertainty.
approv_in_adv	C (B)	Pre-approved or not.	Prior positive screening.
rate_of_interest	N	Interest rate (0–8%).	Risk-based pricing signal.
Interest_rate_spread	N	Spread over market rate.	Risk premium charged.
Upfront_charges	N	Origination fees (0–60K).	Pricing/risk component.
Neg_ammortization	C (B)	Negative amortization allowed.	Debt can grow over time.
interest_only	C (B)	Interest-only initial period.	Defers principal, fragility.
lump_sum_payment	C (B)	Balloon payment at maturity.	Risk concentrated at end.
submission_of_application	C (B)	Direct vs. via intermediary.	Channel screening quality.
property_value	N	Appraised collateral value.	Recovery upon default.
construction_type	C (B)	Site-built vs. manufactured.	Collateral durability.
occupancy_type	C	Primary, secondary, investment.	Lower default on primary homes.
Secured_by	C (B)	Collateral: home or land.	Land is less liquid.
Security_Type	C (B)	Direct vs. indirect security.	Strength of lender's claim.
total_units	C (O)	Housing units (1U–4U).	Multi-unit = investment-driven.
LTV	N	Loan-to-value ratio (%).	Borrower equity in collateral.
Credit_Score	N	Bureau score (500–900).	Past credit behaviour.
credit_type	C	Applicant's bureau (4 bureaus).	Coverage/scale differences.
co-applicant_credit_type	C (B)	Co-applicant's bureau.	As above.
Credit_Worthiness†	C (B)	Internal risk class (l1/l2).	Lender's own assessment.
open_credit	C (B)	Other open credit lines.	Competing debt burden.
dtir1	N	Debt-to-income ratio (5–61%).	Repayment sustainability.
Status	C (B)	Target: 1 = default, 0 = repaid.	Target variable.

3. Pipeline and Experiments

3.1 Preprocessing

The preprocessing stage operates in two phases: a set of *global* cleaning operations applied once to the raw data, and a set of *learned* transformations (imputation, scaling, encoding) that are embedded in `Pipeline` and therefore re-fitted inside every cross-validation fold to avoid any data leakage.

Three operations are applied to the full dataset because they do not estimate any parameter from the data. The missing values of `loan_limit` (2.25%) are mapped to an explicit `unknown` category: for a categorical variable, "not recorded" is itself a legitimate level, and this choice preserves the information carried by the missingness without inventing a value. Then records with missing values in five low-missingness features are dropped: `age`, `loan_purpose`, `Neg_ammortization`, `term` and `submission_of_application`. The remaining missing values `dtir1` (16.12%), `income` (6.04%) and `approv_in_adv` (0.61%) are imputed inside the pipeline: median imputation for numerical features and most-frequent imputation for categorical ones. Numerical features are then standardized with `StandardScaler`, which is required by the scale-sensitive Logistic Regression and harmless for tree-based models; categorical features are one-hot encoded with `handle_unknown="ignore"`, so that categories possibly absent from a training fold do not break prediction on the corresponding validation fold. These steps are assembled in a `ColumnTransformer` placed as the first stage of every model pipeline.
Only for the Naive Bayes model that assumes normal distribution we applied to income a log1p transformation, so that it doesn't violate the assumptions.

3.2 Feature Selection and Engineering

Five features are removed because their availability pattern depends on the target.
`Interest_rate_spread`, `rate_of_interest` and `Upfront_charges` are missing for essentially the entire default class and almost never missing for repaid loans. `property_value` and `LTV` follow an attenuated version of the same pattern (41.2% missing among defaults vs. 0% among non-defaults) and are removed for the same reason.
`credit_type` (the bureau providing the applicant's score), is a leakage source of a different form, detected during model development rather than in the initial screening: in early experiments that retained this feature, the explainability analysis of revealed a suspiciously dominant importance of its one-hot columns, one category becomes a shortcut for the models.
`construction_type` and `Security_Type` are perfectly collinear with `secured_by` and are hence dropped. `ID` and `year` (constant, 2019) carry no signal by construction.

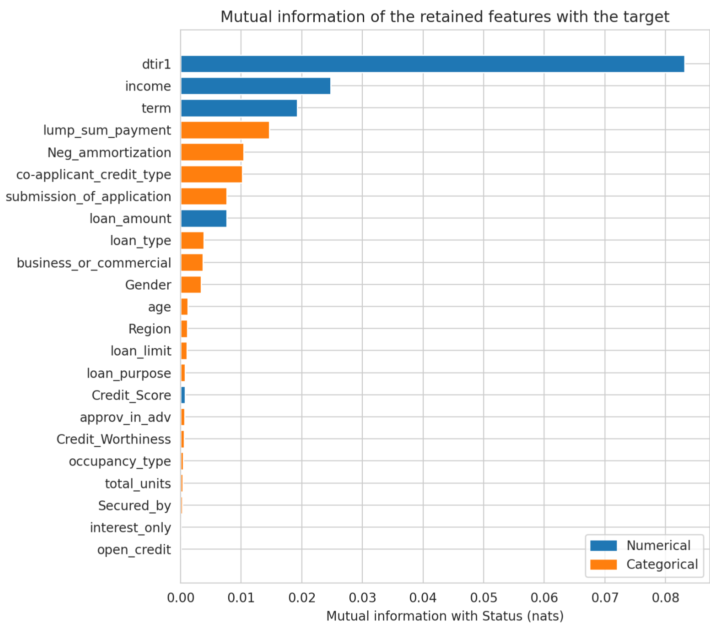

Figure 3. Feature importance scores.

3.3 Model Selection and Cross-Validation

Six classifiers were compared: *Logistic Regression* (linear, interpretable baseline), *Gaussian Naive Bayes* (probabilistic baseline with strong independence assumptions, included to test whether such assumptions hold in the credit domain), *Decision Tree* (non-linear, axis-aligned interactions, fully interpretable), *Random Forest* (bagging ensemble, variance reduction), *AdaBoost* on decision stumps and *Gradient Boosting* (sequential bias-reduction ensembles). This range allows attributing performance gaps to specific modelling capabilities: linearity (LR vs. trees), feature independence (NB vs. all), and ensembling (single tree vs. forests/boosting).

Table 3. Comparison of candidate models. The validation metric is the ROC-AUC used by the inner search to select the configuration; the test metric is the outer-fold ROC-AUC (mean $\pm$ std across the 5 outer folds).

Model	HP tuning	CV strategy	Val. metric	Test metric (AUC)
Logistic Regression	Random Search (30 iter) + local grid	Nested 5×3 stratified	ROC-AUC	0.736 ± 0.002
Naive Bayes (Gaussian)	Local grid (5 iter)	Nested 5×3 stratified	ROC-AUC	0.681 ± 0.003
Decision Tree	Random Search (30 iter) + local grid	Nested 5×3 stratified	ROC-AUC	0.855 ± 0.002
Random Forest	Random Search (15 iter) + local grid	Nested 5×3 stratified	ROC-AUC	0.865 ± 0.001
AdaBoost (stumps)	Random Search (10 iter) + local grid	Nested 5×3 stratified	ROC-AUC	0.831 ± 0.002
Gradient Boosting	Random Search (10 iter) + local grid	Nested 5×3 stratified	ROC-AUC	0.861 ± 0.001

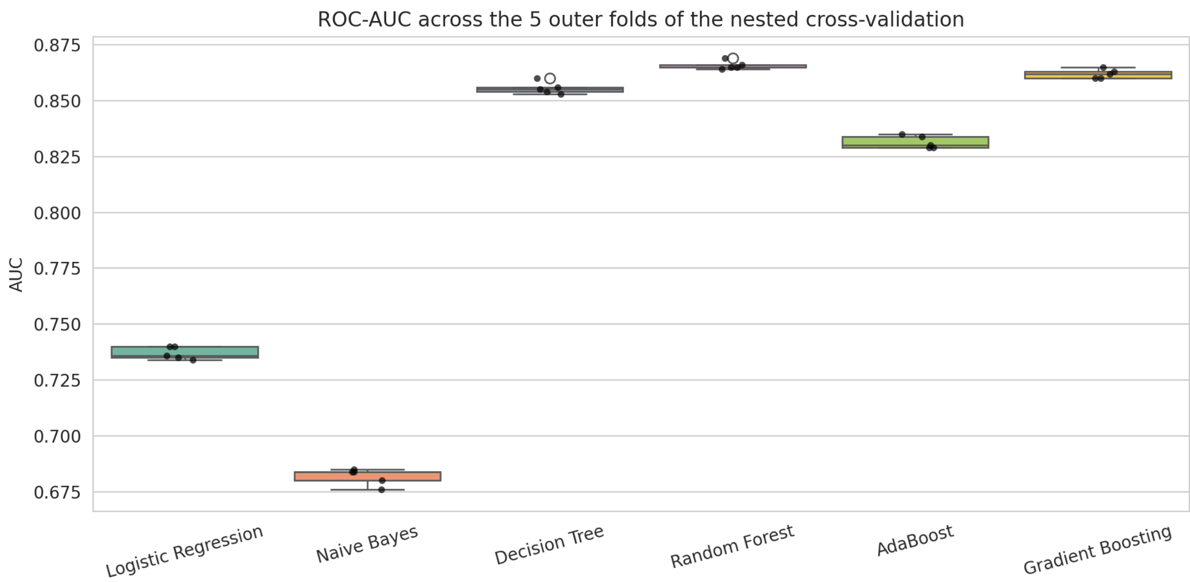

Figure 4. Cross-validated scores by model.

Model comparison uses nested CV: a 5-fold stratified outer loop estimates generalization performance, while a 3-fold stratified inner loop, run independently within each outer training partition, selects hyperparameters. The inner search combines a `RandomizedSearchCV` over a broad space with a subsequent local grid refinement around the best random configuration (function `build_local_grid`, cell 11), using ROC-AUC as the selection metric. Crucially, the object being cross-validated is the entire pipeline: imputers, scaler and encoder are re-fitted on each inner-fold training partition, so no statistic ever leaks from validation data. Stratification preserves the 75/25 class ratio in every fold, which keeps the metrics comparable across folds for an imbalanced problem.

3.4 Hyperparameter Optimization

A two stage strategy is employed, a *random search* over a wide space identifies promising region, and a *grid search* around this region refines it. This reduce the computations needed while covering wide parameter configuration. ROC-AUC is the selection metric used in both stages.

The search spaces are: Logistic Regression — $C \sim \mathrm{loguniform}(10^{-3}, 10^{2})$, solver $\in$ {lbfgs, newton-cg, saga, liblinear}, class weight; Decision Tree — max depth $\in \{3, 5, 8, 10, 15, \mathrm{None}\}$, min samples per leaf $\in \{1, 5, 10, 20, 50\}$, criterion $\in$ {gini, entropy}, class weight; Random Forest — 50–100 estimators, max depth $\in \{10, \mathrm{None}\}$, min samples per leaf $\in \{10, 20\}$, max features $\in \{\sqrt{p}, \log_2 p, 0.3p\}$, class weight; AdaBoost — 50–200 stumps, learning rate $\in [0.01, 1.0]$; Gradient Boosting — 100–300 estimators, learning rate $\in \{0.01, 0.05, 0.1\}$, max depth $\in \{2, 3, 4\}$, min samples per leaf $\in \{20, 50, 100\}$, subsample $\in \{0.6, 0.8, 1.0\}$, max features.

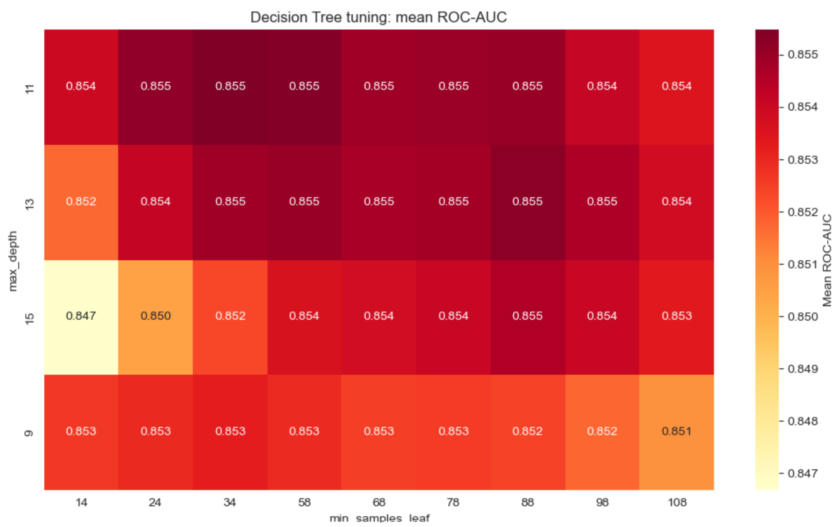

Figure 5. Hyperparameter optimization results.

3.5 Statistical Tests

The three best models, (Random forest, Decision tree, Gradient boost), are compared with the Wilcoxon test on the ROC-AUC of the five paired outer folds. The results are RF vs. DT $p = 0.0625$; RF vs. GB $p = 0.0625$; DT vs. GB $p = 0.0625$. None of the comparison reach significance at $\alpha = 0.05$; indeed, with only $n = 5$ paired observations the smallest attainable two-sided p-value is $2/2^5 = 0.0625$, so the test is structurally underpowered to certify differences of this magnitude.

This non-significance is itself the decisive input for model selection: if Random Forest's nominal advantage ($+0.018$ F1, $+0.010$ AUC over the single tree) is not statistically reliable, the principle of parsimony favours the simplest model in the equivalence class. The Decision Tree is therefore selected as the final model: it trains and predicts orders of magnitude faster than the ensembles, and—decisively for a credit-risk application subject to explainability requirements its decisions can be audited as explicit rule paths, whereas a forest of hundreds of trees cannot.

3.6 Final Model Training

After model-family selection, the final Decision Tree is re-tuned and re-trained on the *entire* cleaned dataset (148,170 records) using the two-stage search of Section 3 with 5-fold stratified CV (cell 59), and refitted with the best configuration (max depth 13, min samples per leaf 88, entropy criterion, no class weighting) on all available data. Retraining on the full dataset is standard practice once the validation protocol is concluded: it lets the deployed model exploit every labelled record, while its expected generalization is *not* read from this final fit but from the unbiased nested-CV estimate of Section 3 ($\mathrm{AUC} = 0.856 \pm 0.003$, balanced accuracy 0.785 ± 0.002, recall 0.679 ± 0.002 on the default class). The closeness between the final 5-fold tuning score (0.8560) and the fully nested estimate (0.8555) confirms that hyperparameter selection introduced no appreciable optimism. The fitted preprocessing pipeline and model are serialized (cell 56) to guarantee that the exact transformation statistics used in training are applied at inference time.

4. Results and Discussion

4.1 Performance Evaluation

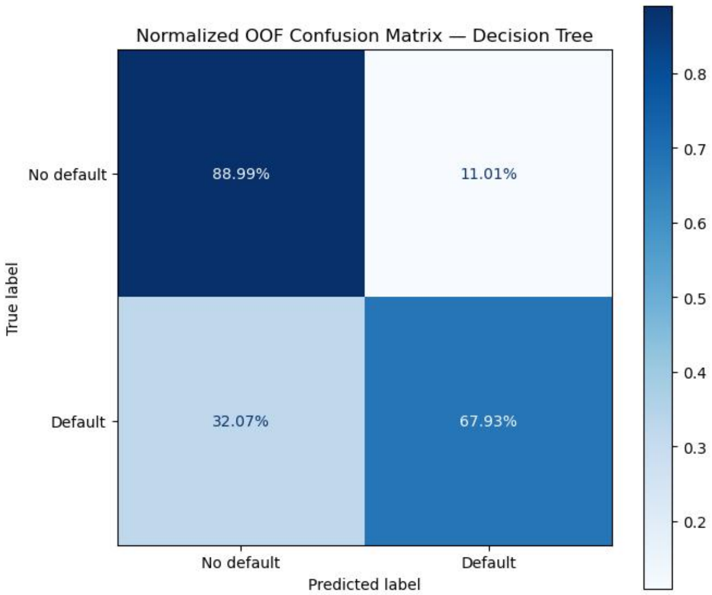

Figure 6. Primary performance result on the test set.

The out of fold metrics for Decision tree are: *Recall = 0.679*, *Precision = 0.679*, *Specificity = 0.889*. These values show how the models has an overall good performance but still struggles in both recognizing all defaults and precision in declaring default. Still the model obtained a balanced result which doesn't favour the bank over the client or viceversa.

The ROC curve confirms that performance of this three model is similar. The rapid ascent of the curve tells us that the model classifies correctly more than it does incorrectly

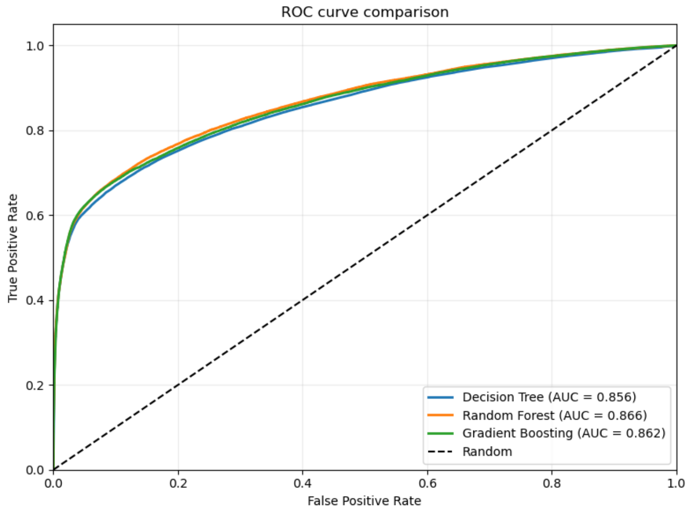

Figure 7. Secondary performance metric on the test set.

4.2 Model Explainability

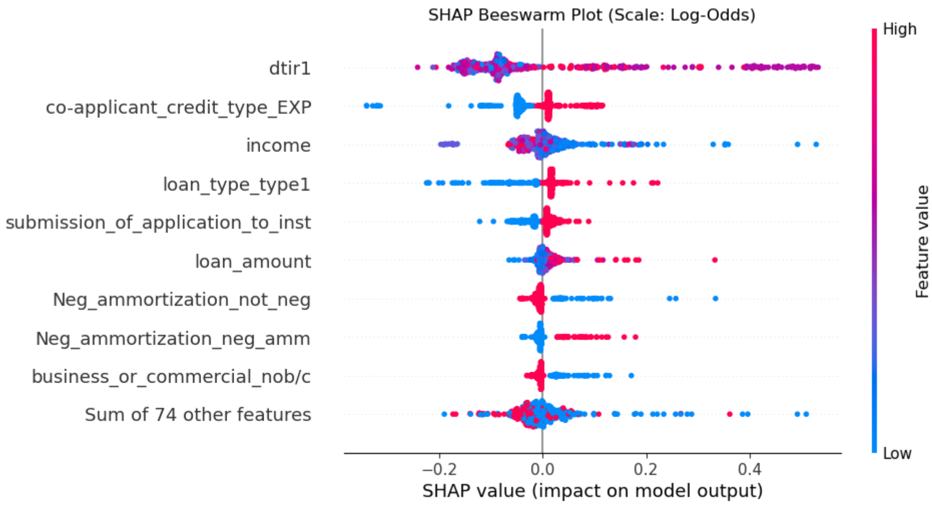

Figure 8. Global explainability: mean $|\mathrm{SHAP}|$ values.

The beeswarm plot shows how feature values importance is quite distinct from high to low. Dtir1, the most important feature is more confused but still it retains good information, moreover together with the income feature.

4.3 Error Analysis

The model's errors are dominated by false negatives (missed defaults) and are not random. Attribution on the misclassified cases shows a single feature, `dtir1` (debt-to-income ratio), driving most of them: a favourable `dtir1` pulls false negatives toward "no default", masking their true risk, while the very same feature in the opposite direction is almost the sole driver of false positives. The model over-relies on it.

`dtir1` is, however, a legitimate primary risk signal, so the errors concentrate on the hard cases where true risk diverges from it—defaulters who look financially healthy on paper. The strong asymmetry (FN $\gg$ FP) and the fact that errors load on the same few features point to a coverage gap (notably the removed `LTV`) rather than noise: recovering these cases would require richer features, not threshold tuning.

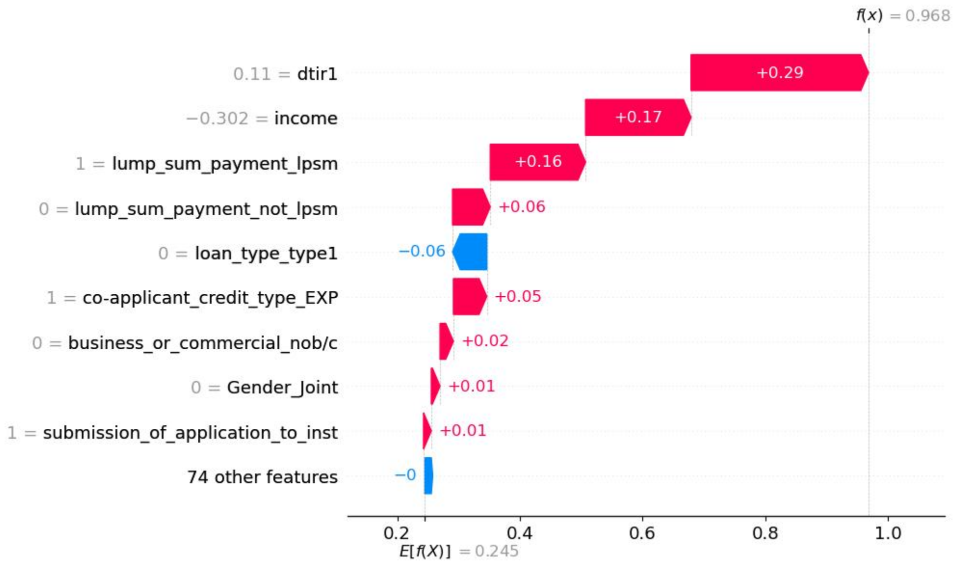

Figure 9. Local explainability: SHAP plot for a representative sample.

5. Conclusions

We addressed bank default prediction as a binary classification task on a large, imbalanced dataset, comparing six model families under a leakage-free nested cross-validation protocol. The three tree-based methods performed best and were statistically indistinguishable (Wilcoxon $p = 0.0625$); by parsimony and interpretability we selected a Decision Tree (max depth 13, min samples per leaf 88) reaching an out-of-fold AUC of 0.856.

Domain impact. The intended use is decision support rather than full automation: the model flags high-risk applications for manual review, and its calibrated probability could feed risk-based pricing and capital-allocation decisions instead of a binary accept/reject. Its low false-positive rate (11.01%) makes it a reliable first-pass screening filter, but the asymmetric cost of errors in credit where a missed default (FN) is far more expensive than a rejected solvent borrower (FP) means it should not be used as a standalone gate.

Limitations. Three limitations qualify these results. First, the dataset carries no timestamp, so temporal drift is undetectable and a statically trained model cannot adapt to macroeconomic shocks such as interest-rate changes. Second, several strong credit-risk predictors had to be discarded: `LTV` and `property_value` for excessive missingness ($> 40\%$), and the pricing features for target leakage; their absence likely caps achievable performance. Third, a single axis-aligned Decision Tree may miss interaction effects that other models capture, and the whole evaluation rests on one dataset with no live deployment test.

Future directions. Future work should collect and restore the most information-rich credit-risk features, and adopt walk-forward (time-aware) cross-validation to simulate deployment and detect drift early. Given that Random Forest was not significantly worse here, calibrated ensembles or gradient boosting on an enriched feature set may yet pay off. Finally, a pilot deployment with human-in-the-loop feedback would allow monitoring live FN/FP rates and recalibrating the decision threshold against realised default costs.